INSPECTION REPORT

ChArUco target · ceramic · 185 × 110 mm

ITEM

Part number	CT-CHA-C-185110-6X6
Description	ChArUco calibration target, ceramic, 185 × 110 mm
Serial number	000311
Lot	L-2611-0382
Datasheet	CT-DS-CHA-C · Rev A
Pattern	ChArUco 7 × 12 · 14 / 10.5 mm DICT_6X6_250
Marker ID range	0–41 (42 markers)
Start ID	0

INSPECTION

Inspection date	2026-10-24
Inspector	JG
Location	per Calib.Tools procedure CT-WI-001
Instrument	Optical CMM · OCMM-01
Last calibration	2026-03 · valid to 2027-03
Temperature	20.4 °C ambient · 20 °C reference

OVERALL RESULT

PASS

All measured features within datasheet tolerance.

9 geometry · 1 flatness · 4 visual
66 corners · U = 1.5 µm (k = 2)

1 Method

All 66 ChArUco corners were measured by comparison against golden sample GS-CHA-C-185110-6X6 (DAkkS-calibrated, certificate [number]) on Optical CMM (OCMM-01); flatness was measured at 25 points. Expanded uncertainty U = 1.5 µm (k = 2), combining golden-sample certificate and comparison uncertainties. Results corrected to 20 °C reference; ambient 20.4 °C.

2 Pattern geometry

ceramic: deviations in µm

FEATURE	NOMINAL (mm)	MEASURED (mm)	DEVIATION (µm)	TOLERANCE (µm)	RESULT
Square pitch X	14.000	14.0008	+0.8	±2	PASS
Square pitch Y	14.000	13.9994	−0.6	±2	PASS
Pattern width	168.000	168.003	+3	±5	PASS
Pattern height	98.000	98.002	+2	±5	PASS
Diagonal 1	194.454	194.457	+3	±7	PASS
Diagonal 2	194.454	194.452	−2	±7	PASS
Orthogonality	90.000°	90.0006°	–	±0.002°	PASS
Max single-feature deviation · n = 66	–	–	1.4	≤ 2	PASS
RMS deviation · information only	–	–	0.6	–	N/A

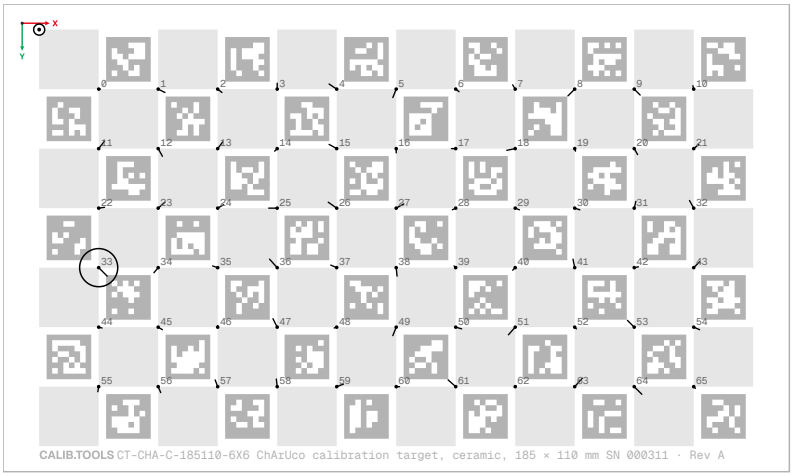
3 Flatness

MEASURED FLATNESS (µm)	TOLERANCE (µm)	METHOD	RESULT
12	≤ 20	25 points, 5 × 5 grid, least-squares plane	PASS

4 Visual

CHECK	CRITERION	RESULT
Edge chamfer	0.5 × 45° on all edges, no chips > 0.3 mm	PASS
Surface defects	No scratches or pinholes in the pattern area	PASS
Print / edge definition	Square edges sharp, no ink bleed at 10× loupe	PASS
Marking legibility	Label line, X / Y arrows and SN legible	PASS

5 Feature map



LEGEND

X → right

Y ↓ down

⊙ Origin (0, 0): top-left outer corner of the pattern (OpenCV frame)

• Measured corner with id, n = 66

—— Deviation vector, magnified × 2000 (1 µm = 2 mm on the board)

◯ Largest deviation: id 33, 1.4 µm

6 Measured coordinates

Complete corner coordinates are supplied in CT-CSV-000311.csv (66 rows; columns id, row, col, x_nom, y_nom, x_meas, y_meas, dx_um, dy_um; mm unless stated). First 10 rows:

id	row	col	x_nom	y_nom	x_meas	y_meas	dx_um	dy_um
0	0	0	14.000	14.000	14.0002	14.0000	+0.2	0.0
1	0	1	28.000	14.000	28.0008	14.0004	+0.8	+0.4
2	0	2	42.000	14.000	42.0005	14.0004	+0.5	+0.4
3	0	3	56.000	14.000	56.0000	13.9993	0.0	-0.7
4	0	4	70.000	14.000	69.9991	13.9994	-0.9	-0.6
5	0	5	84.000	14.000	83.9996	14.0010	-0.4	+1.0
6	0	6	98.000	14.000	98.0004	14.0003	+0.4	+0.3
7	0	7	112.000	14.000	111.9997	13.9995	-0.3	-0.5
8	0	8	126.000	14.000	125.9992	14.0008	-0.8	+0.8
9	0	9	140.000	14.000	140.0007	14.0007	+0.7	+0.7

7 Statement

The item identified above meets the specifications of datasheet CT-DS-CHA-C rev A at the time of inspection. This is an inspection report issued by Calib.Tools; it is not an accredited calibration certificate. An ILAC-MRA certificate issued by an accredited third-party laboratory is available as SVC-X-CERT-ILAC.

8 Traceability

Instrument	OCMM-01 · certificate K-2026-03-0141 · valid to 2027-03
Reference standard	Golden sample GS-CHA-C-185110-6X6 · DAkkS certificate [number] · valid to [date]
Raw data	Raw data retained for 10 years under this serial number.

SIGNED

JG

JG · Inspector · 2026-10-24
Printed signature

CALIB.TOOLS SN 000311 INSPECTED 2026-10-24 JG DIMENSIONAL INSPECTION

calib.tools/verify/000311